

# The Lizard Of Oz

**Future software will go beyond the intelligence of the wizards who will create them**

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**W**e speak in awe of the Artificially Intelligent being. People have come and gone thinking that they would see the ultimate Computer in their lifetime—one that's indistinguishable from a human being. In the past and even today, many researchers have tried to build such a computer.

Alan M Turing, in 1950, proposed the "imitation game" to test the intelligence of a machine. The Turing Test for Artificial Intelligence basically states that if you were chatting with a human and a machine and couldn't tell the difference, then the machine is intelligent; it has "reached AI."

To achieve the technological Nirvana of an intelligent computer, AI researchers use two approaches. The first approach—the top-down approach—is the creation of an all-knowing machine. Machines are great at storing knowledge, but they are awful at little things like basic common sense, so rather than programming the machine with all the

knowledge in the world, program the computer with all the common sense in the world, and let it loose upon civilisation. This is what Douglas Lenat, CEO of Cycorp, is using for his robot Cyc (pronounced "psych"). There are problems with this approach, though.

First, there is the horrifyingly large size of the world's collective common sense. Second, most people would not be able to tell the difference between such a machine and a human, and it would pass the Turing Test. However, there will always be the odd person who would figure out it was a machine he was talking to, and it would then be easy for him to poke holes in the software.

It's nice to know a lot, but realistically, you can't know everything. To be regarded as human, software cannot take shortcuts that we aren't privy to—it will have to go through life just as we do. This is the reasoning Rodney Brooks of MIT uses for his

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